Stat 6541 Midterm 1: Reasoning Behind the Solutions

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These notes expand the answer key: for each part, a *Sketch* of why the argument has the shape it does, a complete *Proof*, the *Supporting details* (which textbook result is used and why it applies), and comments *On the key* where the key is terse, has a gap, or has a typo. Result numbers refer to Chapter 1 and Appendix A of the course text. Throughout, $\subset$ means $\subseteq$, and "countable" means finite or countably infinite. (Definition A.1.11 literally reserves "countable" for sets in bijection with $\mathbb{N}$, but Example 1.1.4 needs the broader reading: $\Omega \in \mathcal{F}_6$ requires $\Omega^c = \emptyset$ to count as countable.)

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5 Summary **10**

1 Three ideas the whole exam runs on

Nearly every part is an instance of one of the following.

(I) Disjointification. Measures are defined through *disjoint* unions (Definition 1.2.1(c)), but the sets one is handed are rarely disjoint. The fix is always the same.

Lemma 1 (Disjointification)**.** Let $\mathcal{A}$ be an algebra on Ω and $A_1, A_2, \ldots \in \mathcal{A}$. Let $B_1 \equiv A_1$ and $B_n \equiv A_n \setminus (A_1 \cup \cdots \cup A_{n-1})$ for $n \geq 2$. Then (i) $B_n \in \mathcal{A}$; (ii) $B_n \subset A_n$; (iii) the B_n are pairwise disjoint; (iv) $\bigcup_{n=1}^{N} B_n = \bigcup_{n=1}^{N} A_n$ for every $N \leq \infty$.

Proof. For (i), $B_n = A_n \cap (A_1 \cup \cdots \cup A_{n-1})^c$, which uses only a finite union, a complement, and a finite intersection, all allowed in an algebra (Definition 1.1.1 and property (c)$'$). Property (ii) is immediate. For (iii), let $m < n$ be given; then $B_m \subset A_m$ while $B_n \subset A_m^c$, so $B_m \cap B_n = \emptyset$. For (iv), ($\subset$) follows from (ii). ($\supset$) Let $\omega \in \bigcup_{n \leq N} A_n$ be given and let $n^* \equiv \min\{n : \omega \in A_n\}$, which exists by the well ordering of $\mathbb{N}$ (Section A.1.2). Then $\omega \in A_{n^*}$ and $\omega \notin A_j$ for $j < n^*$, i.e. $\omega \in B_{n^*}$ with $n^* \leq N$. //

The idea is "charge each point to the first set that contains it." The text uses the same trick in the proofs of Proposition 1.2.1 and Theorem 1.3.2, and its cousin $A_n \equiv \bigcup_{j \leq n} B_j$ (turn an arbitrary union into an *increasing* one) in Proposition 1.1.1.

(II) Identifying a generated class by two containments. $\sigma\langle\mathcal{C}\rangle$ is defined from the outside, as the intersection of all σ-algebras containing $\mathcal{C}$ (Definition 1.1.3); likewise for the generated algebra $\mathcal{A}\langle\mathcal{C}\rangle$. To show such a class equals a concrete candidate $\mathcal{G}$, we prove

- ($\subset$): $\mathcal{G}$ is itself a (σ-)algebra containing $\mathcal{C}$, so the smallest one is inside it;
- ($\supset$): every (σ-)algebra containing $\mathcal{C}$ is *forced* to contain each element of $\mathcal{G}$.

The candidate $\mathcal{G}$ is found by asking what the closure rules force in, and stopping when the collection is closed.

(III) Check on generators, then invoke uniqueness. To prove an identity $\nu_1(A) = \nu_2(A)$ for *all* Borel A, one cannot inspect a general Borel set. Instead: verify the identity on intervals by computation, check that both sides are measures, and let a uniqueness theorem (Theorem 1.2.4 or 1.3.6) carry the identity to $\mathcal{B}(\mathbb{R})$. This is exactly how the text gets reflection and translation invariance of Lebesgue measure (Section 1.3.2, item (iv)).

2 Problem 1

Let $(\Omega, \mathcal{F}, \mu)$ be a measure space.

(a)

Problem. *For $A, B \in \mathcal{F}$, if $\mu(A \setminus B) = 0$ and $\mu(B \setminus A) = 0$, show that $\mu(A) = \mu(B)$.*

Sketch. A and B share the core $A \cap B$, and each differs from that core by a null piece. So rather than compare A to B directly, we compare each to $A \cap B$. Everything is done by *adding* measures of disjoint pieces, never subtracting.

Proof. We have $A = (A \setminus B) \cup (A \cap B)$, and the two pieces are disjoint members of $\mathcal{F}$. By finite additivity,

$$\mu(A) = \mu(A \setminus B) + \mu(A \cap B) = \mu(A \cap B).$$

By the same argument with the roles of A and B exchanged, $\mu(B) = \mu(B \setminus A) + \mu(A \cap B) = \mu(A \cap B)$. Hence $\mu(A) = \mu(B)$. //

Supporting details. The pieces lie in $\mathcal{F}$ because $A \setminus B = A \cap B^c$ and $\mathcal{F}$ is closed under complements and finite intersections. Finite additivity is countable additivity (Definition 1.2.1(c)) with $A_3 = A_4 = \cdots = \emptyset$, as in the first line of the proof of Proposition 1.2.1.

The tempting shortcut "$\mu(A) - \mu(B) = \mu(A \setminus B) - \mu(B \setminus A) = 0$" is *not* valid in general: if $\mu(A \cap B) = \infty$ it reads $\infty - \infty$, which is undefined (Definition A.2.2). The proof above has no such restriction, and the conclusion includes the case $\mu(A) = \mu(B) = \infty$.

The hypothesis says precisely that $\mu(A \triangle B) = 0$, since $A \triangle B$ is the disjoint union of $A \setminus B$ and $B \setminus A$. So (a) says: sets that differ by a null set have the same measure. This is what makes $\tilde{\mu}$ well defined in the completion theorem (Theorem 1.4.1(ii)). The converse is false: $(0, 1]$ and $(1, 2]$ have the same Lebesgue measure but are disjoint.

(b)

Problem. *Let $A_i \in \mathcal{F}$ with $\mu(A_i) = 0$ for all $i \geq 1$. Show that $\mu\big(\bigcup_{i=1}^{\infty} A_i\big) = 0$.*

Sketch. The A_i may overlap, so countable additivity does not apply directly. The tool built for overlapping sets is countable *sub*additivity, which bounds the measure of the union above by 0; nonnegativity bounds it below.

Proof. By Definition 1.2.1(a) and countable subadditivity (Proposition 1.2.3(ii)),

$$0 \;\leq\; \mu\Big(\bigcup_{i=1}^{\infty} A_i\Big) \;\leq\; \sum_{i=1}^{\infty} \mu(A_i) \;=\; 0.$$

//

Supporting details. Proposition 1.2.3(ii) is itself an application of Lemma 1, and it is worth seeing the argument specialized to this problem. With B_n as in the lemma, $B_n \subset A_n$ gives $\mu(B_n) \leq \mu(A_n) = 0$ by monotonicity (Proposition 1.2.2(i)), and then

$$\mu\Big(\bigcup_n A_n\Big) = \mu\Big(\bigcup_n B_n\Big) = \sum_n \mu(B_n) = 0$$

by countable additivity over the disjoint B_n. (The text's own proof goes through m.c.f.b. and finite subadditivity instead.)

Countability is essential. $[0,1] = \bigcup_{x\in[0,1]}\{x\}$ is a union of Lebesgue-null sets with $m([0,1]) = 1$; no contradiction arises because the union is uncountable and measures say nothing about those. The standard consequence is that every countable set is Lebesgue-null (Section 1.3.2, item (ii)), e.g. $m(\mathbb{Q}) = 0$. We use (b) again in Problem 3(d).

On the key. The key writes "since probability is non-negative." Here μ is an arbitrary measure, possibly infinite; the property being used is Definition 1.2.1(a).

(c)

Problem. *Let λ be a finitely additive set function on a σ-algebra $\mathcal{F}$ with $\lambda(\emptyset) = 0$. If $\lambda\big(\bigcup_{i=1}^{\infty} A_i\big) \leq \sum_{i=1}^{\infty} \lambda(A_i)$ for all sequences of disjoint sets in $\mathcal{F}$, prove λ is a measure.*

Sketch. By Definition 1.2.1 we need (a) $\lambda \geq 0$, (b) $\lambda(\emptyset) = 0$, and (c) countable additivity. Countable additivity is two inequalities. The direction "$\leq$" is the hypothesis. The direction "$\geq$" comes for free from finite additivity: every finite partial union sits inside the full union, so by monotonicity the full union has at least the mass of every partial sum, hence at least the mass of the whole series.

Proof. We take λ to have values in $[0,\infty]$ (see *On the key*), so (a) holds, and (b) is given. We first note that λ is monotone: for $A \subset B$ in $\mathcal{F}$, $B = A \cup (B \setminus A)$ is a disjoint union, so $\lambda(B) = \lambda(A) + \lambda(B \setminus A) \geq \lambda(A)$.

Let $A_1, A_2, \ldots \in \mathcal{F}$ be disjoint, and let $N \in \mathbb{N}$ be given. Since $\bigcup_{i=1}^{N} A_i \subset \bigcup_{i=1}^{\infty} A_i$, monotonicity and finite additivity give

$$\lambda\Big(\bigcup_{i=1}^{\infty} A_i\Big) \;\geq\; \lambda\Big(\bigcup_{i=1}^{N} A_i\Big) \;=\; \sum_{i=1}^{N} \lambda(A_i).$$

The left side does not depend on N, so it is an upper bound for every partial sum. The partial sums are nondecreasing (the terms are nonnegative) and their supremum is $\sum_{i=1}^{\infty} \lambda(A_i) \in [0,\infty]$. Hence

$$\lambda\Big(\bigcup_{i=1}^{\infty} A_i\Big) \;\geq\; \sum_{i=1}^{\infty} \lambda(A_i).$$

Combining this with the hypothesis gives equality, which is (c). //

Supporting details. If "finitely additive" is read as additivity for *two* disjoint sets (as in (iii)$'_a$ of Proposition 1.2.1), additivity for N sets follows by induction: $\bigcup_{i\leq N} A_i$ and A_{N+1} are disjoint. The σ-algebra hypothesis is what guarantees $\bigcup_{i\geq 1} A_i \in \mathcal{F}$, so that λ of it is defined.

This result is a companion to Proposition 1.2.1, which says

$$\text{measure} \iff \text{finitely additive} + \text{m.c.f.b.}$$

Part (c) says

$$\text{measure} \iff \text{finitely additive} + \text{countably subadditive (on disjoint sequences).}$$

In both, finite additivity supplies "$\geq$" and the extra condition supplies "$\leq$." The same two-sided pattern closes the proof of Theorem 1.3.2(ii): there $\mu^*(A_1) \geq \sum_{i\leq n} \mu^*(B_i)$ for all n comes from the finite stage, and the opposite inequality from subadditivity of the outer measure.

The hypothesis is not automatic. On the finite–cofinite algebra of $\mathbb{N}$ (Example 1.1.3), let $\lambda(A) = 0$ if A is finite and 1 if A^c is finite. This is finitely additive, but for $A_i = \{i\}$, $\lambda(\bigcup_i A_i) = \lambda(\mathbb{N}) = 1 > 0 = \sum_i \lambda(A_i)$. So "$\geq$" holds, as it must, and "$\leq$" fails. (That example lives on an algebra; examples on σ-algebras exist but are not constructive.)

On the key. The key says "Since $\lambda \geq 0$," but the problem statement never says so, and the text's "set function" is only extended-real valued (Section 1.2). Nonnegativity cannot be derived: $\lambda(A) \equiv -I_A(0)$ on $\mathcal{B}(\mathbb{R})$ is countably additive with $\lambda(\emptyset) = 0$, so it satisfies every stated hypothesis, and it is not a measure. So $\lambda : \mathcal{F} \to [0, \infty]$ has to be read as part of the setup, as in the statement of Proposition 1.2.1. On an exam, say so in one line.

3 Problem 2

Let Ω be an uncountable set and $\mathcal{A}$ an algebra on Ω.

(a)

Problem. *Prove that if $\mathcal{A}$ is closed under countable disjoint unions, then $\mathcal{A}$ is a σ-algebra.*

Sketch. An algebra already satisfies (a)–(c) of Definition 1.1.1, so only (d) of Definition 1.1.2 is missing: closure under *arbitrary* countable unions. The hypothesis covers only disjoint ones. Lemma 1 rewrites an arbitrary countable union as a disjoint one using finite operations, which an algebra permits.

Proof. Let $A_1, A_2, \ldots \in \mathcal{A}$ be given, and let B_n be as in Lemma 1. By the lemma, each $B_n \in \mathcal{A}$, the B_n are pairwise disjoint, and $\bigcup_{n\geq 1} A_n = \bigcup_{n\geq 1} B_n$. By hypothesis $\bigcup_{n\geq 1} B_n \in \mathcal{A}$. Hence $\bigcup_{n\geq 1} A_n \in \mathcal{A}$, and $\mathcal{A}$ is a σ-algebra. //

Supporting details. The one place the algebra structure is used is $B_n \in \mathcal{A}$: it needs the finite union $A_1 \cup \cdots \cup A_{n-1}$, a complement, and an intersection. Compare Proposition 1.1.1, which reaches the same conclusion from closure under *increasing* unions. Together they give three equivalent ways to upgrade an algebra to a σ-algebra: closure under countable unions that are arbitrary, increasing, or disjoint. The uncountability of Ω plays no role in (a); it matters only in (b) and (c).

(b)

Problem. *Let $\mathcal{C} = \{\{x\} : x \in \Omega\}$. Describe the algebra $\mathcal{A}^* \equiv \mathcal{A}\langle\mathcal{C}\rangle$, the smallest algebra generated by $\mathcal{C}$.*

Sketch. We use idea (II). An algebra containing the singletons must contain their finite unions (all finite sets) and then the complements of those (all cofinite sets). Is anything else forced? No: the finite and cofinite sets already form an algebra, so the process stops. The claim is

$$\mathcal{A}^* = \mathcal{F} \equiv \{E \subset \Omega : E \text{ is finite or } E^c \text{ is finite}\}.$$

Here $\mathcal{A}\langle\mathcal{C}\rangle$ is the intersection of all algebras containing $\mathcal{C}$; this is well defined because $\mathcal{P}(\Omega)$ is one such algebra, and an intersection of algebras is an algebra (the argument of Problem 1.4).

Proof. ($\supset$) Let $\mathcal{G}$ be an algebra containing $\mathcal{C}$ and let $E \in \mathcal{F}$ be given.

Case 1: E is finite. Then $E = \{x_1\} \cup \cdots \cup \{x_k\}$ is a finite union of members of $\mathcal{C} \subset \mathcal{G}$, so $E \in \mathcal{G}$ (if $E = \emptyset$, then $E = \Omega^c \in \mathcal{G}$).

Case 2: E^c is finite. Then $E^c \in \mathcal{G}$ by Case 1, so $E = (E^c)^c \in \mathcal{G}$.

In either case $E \in \mathcal{G}$. Taking $\mathcal{G} = \mathcal{A}^*$ gives $\mathcal{F} \subset \mathcal{A}^*$.

($\subset$) We check that $\mathcal{F}$ is an algebra (Definition 1.1.1). (a) $\Omega^c = \emptyset$ is finite, so $\Omega \in \mathcal{F}$. (b) The definition of $\mathcal{F}$ is symmetric in E and E^c, so $\mathcal{F}$ is closed under complements. (c) Let $A, B \in \mathcal{F}$ be given. If both are finite, so is $A \cup B$. Otherwise one of them, say A, has finite complement, and

then $(A \cup B)^c = A^c \cap B^c \subset A^c$ is finite. In either case $A \cup B \in \mathcal{F}$. Since also $\mathcal{C} \subset \mathcal{F}$, and $\mathcal{A}^*$ is the smallest algebra containing $\mathcal{C}$, we get $\mathcal{A}^* \subset \mathcal{F}$.

Combining both containments, we conclude $\mathcal{A}^* = \mathcal{F}$. //

Supporting details. $\mathcal{F}$ is the class $\mathcal{F}_5$ of Example 1.1.3, where the algebra property is verified through intersections, (c)′, instead of unions. That example also shows $\mathcal{F}$ is *not* a σ-algebra when Ω is infinite: a countably infinite $E \subset \Omega$ is a countable union of singletons, but here E is infinite and E^c is uncountable, so $E \notin \mathcal{F}$. This is why the answer to (c) must be larger.

On the key. In the direction $\mathcal{A}^* \subset \mathcal{F}$, the key says only that $\mathcal{C} \subset \mathcal{F}$ and $\mathcal{A}^*$ is smallest. Minimality applies only among *algebras*, so the step silently uses that $\mathcal{F}$ is an algebra. That verification is the actual content of this direction.

(c)

Problem. *Describe the σ-algebra $\sigma\langle\mathcal{C}\rangle$ generated by $\mathcal{C}$.*

Sketch. Repeat (b) with "finite" replaced by "countable": countable unions of singletons give all countable sets, complements give all cocountable sets, and that collection is closed under countable unions. The claim is

$$\sigma\langle\mathcal{C}\rangle = \mathcal{G} \equiv \{E \subset \Omega : E \text{ is countable or } E^c \text{ is countable}\}.$$

Proof. ($\supset$) Let $\mathcal{H}$ be a σ-algebra containing $\mathcal{C}$ and let $E \in \mathcal{G}$ be given. If E is countable, then $E = \bigcup_{x\in E}\{x\}$ is a countable union of members of $\mathcal{H}$, so $E \in \mathcal{H}$ (if $E = \emptyset$, then $E = \Omega^c \in \mathcal{H}$). If E^c is countable, then $E^c \in \mathcal{H}$ and so $E \in \mathcal{H}$. Hence $\mathcal{G} \subset \mathcal{H}$. Taking $\mathcal{H} = \sigma\langle\mathcal{C}\rangle$ gives $\mathcal{G} \subset \sigma\langle\mathcal{C}\rangle$.

($\subset$) We check that $\mathcal{G}$ is a σ-algebra. As in (b), $\Omega \in \mathcal{G}$ and $\mathcal{G}$ is closed under complements. Let $A_1, A_2, \ldots \in \mathcal{G}$ be given.

Case 1: every A_n is countable. Then $\bigcup_n A_n$ is a countable union of countable sets, hence countable (Problem A.6).

Case 2: some A_{n_0} has countable complement. Then $\left(\bigcup_n A_n\right)^c = \bigcap_n A_n^c \subset A_{n_0}^c$ is countable.

In either case $\bigcup_n A_n \in \mathcal{G}$. Since $\mathcal{C} \subset \mathcal{G}$ and $\sigma\langle\mathcal{C}\rangle$ is the smallest σ-algebra containing $\mathcal{C}$, $\sigma\langle\mathcal{C}\rangle \subset \mathcal{G}$.

Combining both containments, we conclude $\sigma\langle\mathcal{C}\rangle = \mathcal{G}$. //

Supporting details. $\mathcal{G}$ is $\mathcal{F}_6$ of Example 1.1.4 (its σ-algebra property is Problem 1.3). There is also a route through (b): since $\mathcal{C} \subset \mathcal{A}\langle\mathcal{C}\rangle \subset \sigma\langle\mathcal{C}\rangle$, we have $\sigma\langle\mathcal{C}\rangle = \sigma\langle\mathcal{A}\langle\mathcal{C}\rangle\rangle = \sigma\langle\mathcal{F}_5\rangle$, and Example 1.1.7 shows $\sigma\langle\mathcal{F}_5\rangle = \mathcal{F}_6$.

This is where "Ω uncountable" earns its place. If Ω were countable, every subset would be countable and the answer would collapse to $\mathcal{P}(\Omega)$. For uncountable Ω the answer is typically much smaller than $\mathcal{P}(\Omega)$: with $\Omega = \mathbb{R}$, the set $[0, \infty)$ is neither countable nor cocountable, so $[0, \infty) \notin \sigma\langle\mathcal{C}\rangle$. In particular singletons do not generate the Borel σ-algebra, which is Problem 1.12. For $\Omega = \mathbb{R}$ the picture is

$$\mathcal{C} \subsetneq \mathcal{A}\langle\mathcal{C}\rangle \subsetneq \sigma\langle\mathcal{C}\rangle \subsetneq \mathcal{B}(\mathbb{R}).$$

On the key. "By a similar logic extended to countable operations" is the proof above. The two facts that need to be said are Case 2 (one cocountable set makes the whole union cocountable) and that a countable union of countable sets is countable.

4 Problem 3

Let $F(x) = x^3$ and let $\mu \equiv \mu_F$ be the Lebesgue–Stieltjes measure on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$ induced by F.

Setup: what μ is and how to compute with it

F is nondecreasing and continuous, so $F(x+) = F(x-) = F(x)$ and $F(\pm\infty) = \pm\infty$. Following Section 1.3.2, let

$$\mathcal{C} \equiv \{(a, b] : -\infty \le a \le b < \infty\} \cup \{(a, \infty) : -\infty \le a < \infty\}$$

be the semialgebra (3.7). By (3.8) and continuity of F,

$$\mu((a, b]) = F(b+) - F(a+) = b^3 - a^3 \quad (-\infty < a \le b < \infty),$$

and every unbounded member of $\mathcal{C}$ has measure ∞. The measure μ is the Carathéodory extension μ_F^* (Theorem 1.3.3) restricted to $\sigma\langle\mathcal{C}\rangle = \mathcal{B}(\mathbb{R})$. Since intervals and singletons are Borel sets, the μ_F^* in parts (c)–(d) is just μ.

Only right-closed intervals come with a formula. The others are derived.

Lemma 2 (Interval formulas)**.** Let G be nondecreasing and right continuous, and $-\infty < a < b < \infty$. Then

$$\mu_G(\{a\}) = G(a) - G(a-), \qquad \mu_G((a, b)) = G(b-) - G(a),$$
$$\mu_G([a, b]) = G(b) - G(a-), \qquad \mu_G([a, b)) = G(b-) - G(a-).$$

In particular, if G is continuous, every singleton is μ_G-null and all four intervals with endpoints $a < b$ have measure $G(b) - G(a)$.

Proof. The sets $(a - \frac{1}{n}, a]$ decrease to $\{a\}$ and $\mu_G((a-1, a]) < \infty$, so by monotone continuity from above (Proposition 1.2.3(i)),

$$\mu_G(\{a\}) = \lim_{n\to\infty} \big[G(a) - G(a - \tfrac{1}{n})\big] = G(a) - G(a-).$$

The rest is finite additivity with finite quantities: $(a, b] = (a, b) \cup \{b\}$ gives $\mu_G((a, b)) = [G(b) - G(a)] - [G(b) - G(b-)]$; $[a, b] = \{a\} \cup (a, b]$ and $[a, b) = \{a\} \cup (a, b)$ give the other two. //

So the rule is: *a closed endpoint picks up the jump of G there; an open endpoint does not.* The singleton formula is item (i) of Section 1.3.2, which reads $\mu_F^*(\{x\}) = F(x+) - F(x-)$ because the text does not assume right continuity; (3.8) uses $F(\cdot+)$ throughout for the same reason. For $F(x) = x^3$ none of this matters.

A heuristic to keep in mind. Because F is differentiable, μ has density $F'(x) = 3x^2$ with respect to Lebesgue measure: $\mu((a, b]) = \int_a^b 3x^2\,dx$. (The general statement $\mu(A) = \int_A 3x^2\,dx$ needs the integration theory that comes after Chapter 1, so it is a guide here and not a proof.) It predicts every answer below: substituting $x = cy$ gives $\int_{cA} 3x^2\,dx = \int_A 3c^2y^2\,c\,dy = c^3\mu(A)$; the density is even, so $\mu(-A) = \mu(A)$; the density is not constant, so μ is not translation invariant; and a single point carries no mass.

(a)

Problem. *Show, using the Uniqueness Theorem, that* $\mu(cA) = c^3\mu(A)$ *for* $c > 0$ *and* $A \in \mathcal{B}(\mathbb{R})$.

Sketch. This is idea (III). Fix $c > 0$ and regard both sides as functions of A: $\nu_1(A) \equiv \mu(cA)$ and $\nu_2(A) \equiv c^3\mu(A)$. On an interval $(a, b]$ the identity is one line of algebra, and it holds because F is homogeneous of degree 3: $F(cx) = c^3F(x)$. To pass from intervals to all Borel sets we need three things the key states quickly: cA is Borel (so ν_1 makes sense), ν_1 and ν_2 are measures, and the common restriction to $\mathcal{C}$ is σ-finite. Then Theorem 1.3.6 applies.

Proof. Let $c > 0$ be given and write $cA \equiv \{cx : x \in A\}$.

We first check the identity on $\mathcal{C}$. Since multiplication by $c > 0$ preserves order, $c(a, b] = (ca, cb]$ for $-\infty < a \le b < \infty$, and

$$\mu(c(a, b]) = (cb)^3 - (ca)^3 = c^3(b^3 - a^3) = c^3\mu((a, b]).$$

For the unbounded members, $c(a, \infty) = (ca, \infty)$ and $c(-\infty, b] = (-\infty, cb]$, so both sides equal ∞ (using $c^3 \cdot \infty = \infty$ for $c > 0$).

Next, $cA \in \mathcal{B}(\mathbb{R})$ for every $A \in \mathcal{B}(\mathbb{R})$. Let $\mathcal{L} \equiv \{A \in \mathcal{B}(\mathbb{R}) : cA \in \mathcal{B}(\mathbb{R})\}$. Because $x \mapsto cx$ is a bijection of $\mathbb{R}$, we have $c\mathbb{R} = \mathbb{R}$, $c(A^c) = (cA)^c$, and $c\big(\bigcup_n A_n\big) = \bigcup_n cA_n$; hence $\mathcal{L}$ is a σ-algebra. By the previous paragraph $\mathcal{C} \subset \mathcal{L}$, so $\mathcal{L} \supset \sigma\langle\mathcal{C}\rangle = \mathcal{B}(\mathbb{R})$.

Now ν_1 is a measure on $\mathcal{B}(\mathbb{R})$: it is nonnegative, $\nu_1(\emptyset) = \mu(\emptyset) = 0$, and if $A_1, A_2, \ldots$ are disjoint Borel sets then $cA_1, cA_2, \ldots$ are disjoint (injectivity), so

$$\nu_1\Big(\bigcup_n A_n\Big) = \mu\Big(\bigcup_n cA_n\Big) = \sum_n \mu(cA_n) = \sum_n \nu_1(A_n).$$

A positive multiple of a measure is a measure, so ν_2 is one too.

Finally, let μ_0 be the restriction of ν_2 to $\mathcal{C}$. It is a measure on the semialgebra $\mathcal{C}$, and it is σ-finite because $\mathbb{R} = \bigcup_{n\in\mathbb{Z}}(n, n+1]$ with $\mu_0((n, n+1]) = c^3[(n+1)^3 - n^3] < \infty$. Both ν_1 and ν_2 are measures on $(\mathbb{R}, \sigma\langle\mathcal{C}\rangle)$ that agree with μ_0 on $\mathcal{C}$. By Theorem 1.3.6, $\nu_1 = \mu_0^* = \nu_2$ on $\mathcal{B}(\mathbb{R})$; that is, $\mu(cA) = c^3\mu(A)$ for all $A \in \mathcal{B}(\mathbb{R})$. //

Supporting details. *Why the proof needs the measure property.* Agreement on intervals says nothing by itself about a set such as the Cantor set. Uniqueness theorems compare *measures*, so the work is in showing that $A \mapsto \mu(cA)$ is one. The reason it is: scaling is a bijection, so it commutes with every set operation. (Once measurable functions are available, $cA = T^{-1}(A)$ for the continuous map $T(x) = x/c$, and $\nu_1 = \mu T^{-1}$ is an induced measure; that is the same argument in one line.)

Why σ-finiteness cannot be dropped. On $\mathcal{B}(\mathbb{R})$ let ν_1 be counting measure and $\nu_2 = 2\nu_1$. Every nonempty $(a, b]$ and every ray is infinite, so $\nu_1 = \nu_2 = \infty$ there and the two measures agree on all of $\mathcal{C}$; but $\nu_1(\{0\}) = 1 \ne 2 = \nu_2(\{0\})$. Their common restriction to $\mathcal{C}$ is not σ-finite, which is what breaks.

The π–λ route. The key phrases the last step as "a π-system that generates $\mathcal{B}(\mathbb{R})$, and two σ-finite measures." That is Theorem 1.2.4 extended as in Problem 1.26(b). In detail: for $n \in \mathbb{N}$ let $\nu_i^{(n)}(A) \equiv \nu_i(A \cap (-n, n])$, $i = 1, 2$. These are finite measures; they agree on the π-system of bounded intervals $(a, b]$ (the intersection of two such intervals is another one, or empty); and they have the same total mass $\nu_i((-n, n])$. Theorem 1.2.4 gives $\nu_1^{(n)} = \nu_2^{(n)}$ on $\mathcal{B}(\mathbb{R})$, and letting $n \to \infty$ with m.c.f.b. gives $\nu_1 = \nu_2$. Either route is fine; behind both is the π–λ theorem (Theorem 1.1.2).

What is really going on. For any nondecreasing right-continuous G, the computation above shows $A \mapsto \mu_G(cA)$ is the Lebesgue–Stieltjes measure of $x \mapsto G(cx)$. For $G(x) = x$ this is $c\,m$,

the scaling rule for Lebesgue measure; for $G(x) = x^3$ it is $c^3\mu$. The exponent is the degree of homogeneity of F. A quick check: $\mu((0,2]) = 8 = 2^3 \cdot 1 = 2^3\mu((0,1])$.

On the key. (i) "Operations of scaling sets preserve the properties of σ-algebras" is the paragraph on $\mathcal{L}$ and the verification that ν_1 is a measure. (ii) "We just established that $\nu_1(A) = \nu_2(A)$ for all $A \in \mathcal{C}$" was only shown for bounded $(a, b]$; the rays need the $\infty = \infty$ check. (iii) The result invoked is the text's Theorem 1.3.6 (or Theorem 1.2.4 with the σ-finite extension); the *extension* theorem, Theorem 1.3.3, is the one named for Carathéodory.

(b)

Problem. *Show that μ is reflection invariant: $\mu(-A) = \mu(A)$.*

Sketch. Same pattern as (a), with $x \mapsto -x$ in place of $x \mapsto cx$. The new feature is that reflection reverses order, so it sends the right-closed interval $(a, b]$ to the *left*-closed interval $[-b, -a)$, which is not in $\mathcal{C}$. Lemma 2 handles this: F is continuous, so endpoints carry no mass. After that, the identity on intervals holds because F is odd.

Proof. Let $-\infty < a < b < \infty$ be given. Then $-(a, b] = \{-x : a < x \leq b\} = [-b, -a)$, and by Lemma 2 with F continuous,

$$\mu(-(a, b]) = \mu([-b, -a)) = F(-a) - F(-b) = -a^3 + b^3 = \mu((a, b]).$$

(For $a = b$ both sides are $\mu(\emptyset) = 0$.) For the unbounded members of $\mathcal{C}$, $-(a, \infty) = (-\infty, -a)$ and $-(-\infty, b] = [-b, \infty)$ each contain bounded intervals of arbitrarily large measure, so by monotonicity both have measure ∞, matching $\mu((a, \infty)) = \mu((-\infty, b]) = \infty$.

Let $\nu_1(A) \equiv \mu(-A)$ and $\nu_2 \equiv \mu$. Since $x \mapsto -x$ is a bijection of $\mathbb{R}$, the argument of (a) applies verbatim: $-A \in \mathcal{B}(\mathbb{R})$ for all $A \in \mathcal{B}(\mathbb{R})$, and ν_1 is a measure on $\mathcal{B}(\mathbb{R})$. We have shown $\nu_1 = \nu_2$ on $\mathcal{C}$, and μ restricted to $\mathcal{C}$ is σ-finite. By Theorem 1.3.6, $\nu_1 = \nu_2$ on $\mathcal{B}(\mathbb{R})$. //

Supporting details. The computation uses two properties of F: it is odd, so $F(-a) - F(-b) = F(b) - F(a)$, and it is continuous, so $[-b, -a)$ and $(-b, -a]$ have the same measure. Oddness is the essential one. For instance $G(x) = e^x$ is continuous but $\mu_G((0,1]) = e - 1 \neq 1 - e^{-1} = \mu_G([-1, 0))$, so μ_G is not reflection invariant.

On "rotation/reflection": in one dimension the only distance-preserving linear maps are $x \mapsto \pm x$, so reflection is all there is. Note that μ is *not* translation invariant, unlike Lebesgue measure (Section 1.3.2, item (iv)): $\mu((0,1]) = 1$ but $\mu((1,2]) = 7$.

Combining (a) and (b): for $c < 0$, $cA = -(|c|A)$, so $\mu(cA) = |c|^3\mu(A)$ for every $c \neq 0$.

On the key. The key's phrase "using the fact that $F(x) = x^3$ is continuous" is Lemma 2: without continuity, $\mu([-b, -a)) = F((-a)-) - F((-b)-)$, and the left limits could differ from the values.

(c)

Problem. *Calculate $\mu_F^*([-2, -1])$.*

Sketch. A closed interval is a right-closed interval plus its left endpoint. The endpoint is null because F is continuous.

Proof. Since $[-2, -1] = \{-2\} \cup (-2, -1]$ is a disjoint union,

$$\mu([-2, -1]) = \mu(\{-2\}) + \mu((-2, -1]) = 0 + \left[(-1)^3 - (-2)^3\right] = -1 + 8 = 7.$$

//

Supporting details. For a general nondecreasing right-continuous G the formula is $\mu_G([a,b]) = G(b) - G(a-)$ (Lemma 2); the key's $F(-1) - F(-2)$ is this with $F(-2-) = F(-2)$. Three checks: the answer is nonnegative, which catches the common sign slip in $(-1)^3 - (-2)^3$; by (b), $\mu([-2,-1]) = \mu([1,2]) = 8 - 1 = 7$; and the density heuristic gives $\int_{-2}^{-1} 3x^2\,dx = 7$. The interval has length 1 but measure 7: μ weights a set by where it sits, not only by how long it is.

(d)

Problem. *Calculate* $\mu_F^*(\{3\})$.

Sketch. The mass of a point is the jump of F there, and F has no jumps.

Proof. By Lemma 2, $\mu(\{3\}) = F(3) - F(3-) = 27 - 27 = 0$. //

Supporting details. A direct argument without the lemma: for every $\epsilon \in (0,1)$, $\{3\} \subset (3-\epsilon, 3]$, so by monotonicity

$$0 \le \mu(\{3\}) \le 27 - (3-\epsilon)^3 \longrightarrow 0 \quad \text{as } \epsilon \downarrow 0.$$

In general the atoms of μ_G are exactly the discontinuities of G. Here there are none, so by Problem 1(b) every countable set is μ-null (e.g. $\mu(\mathbb{Q}) = 0$), and by Problem 1(a) changing a set by countably many points does not change its measure. That is the reason open, closed, and half-open intervals were interchangeable in (b) and (c).

On the key. The key's last line evaluates at $x = 4$ ($4^3 - 4^3 = 0$); the question asks about $\{3\}$, so it should read $3^3 - 3^3 = 0$. Also, "strictly continuous" is not a standard term; plain continuity at the point is what is used.

5 Summary

Places where the key needs a patch.

- 1(b): "probability" should be "measure."
- 1(c): $\lambda \ge 0$ is used but not in the problem statement, and it cannot be derived from what is stated.
- 2(b): the containment $\mathcal{A}^* \subset \mathcal{F}$ needs $\mathcal{F}$ to be an algebra (Example 1.1.3).
- 3(a): $cA \in \mathcal{B}(\mathbb{R})$ and "ν_1 is a measure" need the bijection argument; agreement on the unbounded members of $\mathcal{C}$ needs a separate line.
- 3(d): the computation is written for $\{4\}$ instead of $\{3\}$. The answer, 0, is unaffected.

Patterns to carry forward.

- To prove an equality of measures of sets, split into disjoint pieces and add; do not subtract unless the quantities are known to be finite.
- Countable additivity is two inequalities: finite additivity plus monotonicity gives "$\ge$"; subadditivity (or continuity from below) gives "$\le$."
- To identify $\mathcal{A}\langle\mathcal{C}\rangle$ or $\sigma\langle\mathcal{C}\rangle$: guess the closure, show the guess is a (σ-)algebra, and show every (σ-)algebra containing $\mathcal{C}$ contains the guess.
- To prove an identity for all Borel sets: intervals, then "both sides are measures," then σ-finiteness and Theorem 1.3.6.
- For Lebesgue–Stieltjes measures: closed endpoints pick up jumps, and a continuous F has no atoms.